Sample Paper 01

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

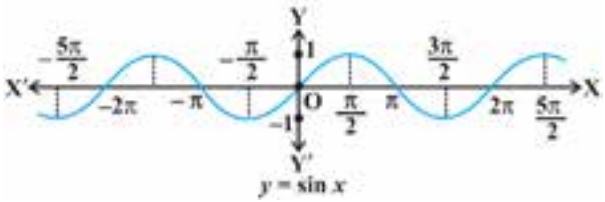
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Multiple Choice Questions each question carries 1 mark.

1. At which point does the graph of $y = \sin x$ intersect the origin in the given image?



(a) $x = -\pi$ (b) $x = 0$

(c) $x = \pi$ (d) $x = 2\pi$

2. If A and B are two independent events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$, then P (neither A nor B) is equal to

(a) $\frac{2}{3}$ (b) $\frac{1}{6}$

(c) $\frac{5}{6}$ (d) $\frac{1}{3}$

3. The distance of the plane $6x - 3y + 2z - 14 = 0$ from the origin is

(a) 2 (b) 1

(c) 14 (d) 8

4. If $P(A \cup B) = 0.83$, $P(A) = 0.3$ and $P(B) = 0.6$, then the events will be

(a) dependent

(b) independent

(c) cannot say anything

(d) None of the above

5. If $\vec{x}$ and $\vec{y}$ are unit vectors and $\vec{x} \cdot \vec{y} = 0$, then

(a) $|\vec{x} + \vec{y}| = 1$ (b) $|\vec{x} + \vec{y}| = \sqrt{3}$

(c) $|\vec{x} + \vec{y}| = 2$ (d) $|\vec{x} + \vec{y}| = \sqrt{2}$

6. $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx$ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

7. The area bounded by the curve $y = \frac{1}{2}x^2$, the X-axis and the ordinate $x = 2$ is

(a) $\frac{1}{3}$ sq unit (b) $\frac{2}{3}$ sq unit

(c) 1 sq unit (d) $\frac{4}{3}$ sq unit

8. The solution of $\frac{dy}{dx} = \frac{ax + g}{by + f}$ represents a circle, when

(a) $a = b$

(b) $a = -b$

(c) $a = -2b$

(d) $a = 2b$

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9. If A and B are two symmetric matrices of same order. Then, the matrix $AB - BA$ is equal to

(a) a symmetric matrix

(b) a skew-symmetric matrix

(c) a null matrix

(d) the identity matrix

10. The family of curves $y = e^{a \sin x}$, where a is an arbitrary constant is represented by the differential equation

(a) $\log y = \tan x \frac{dy}{dx}$ (b) $y \log y = \tan x \frac{dy}{dx}$

(c) $y \log y = \sin \frac{dy}{dx}$ (d) $\log y = \cos x \frac{dy}{dx}$

11. The value of $\int_{-2}^{2} (x \cos x + \sin x + 1)\, dx$ is

(a) 2 (b) 0

(c) -2 (d) 4

12. $\int \frac{\sin 2x}{\sin^2 x + 2\cos^2 x} dx$ is equal to

(a) $-\log(1 + \sin^2 x) + C$

(b) $\log(1 + \cos^2 x) + C$

(c) $-\log(1 + \cos^2 x) + C$

(d) $\log(1 + \tan^2 x) + C$

13. If $|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = 144$ and $|\vec{a}| = 4$, then $|\vec{b}|$ is equal to

(a) 16 (b) 8

(c) 3 (d) 12

14. If A and B are two equivalence relations defined on set C, then

(a) $A \cap B$ is an equivalence relation

(b) $A \cap B$ is not an equivalence relation

(c) $A \cap B$ is an equivalence relation

(d) $A \cap B$ is not an equivalence relation

15. In Fig., the shaded region is bounded by which of the following conditions?

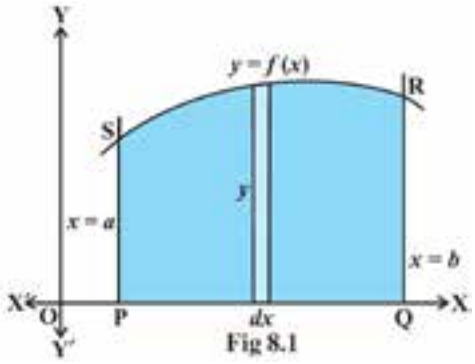


Fig 8.1

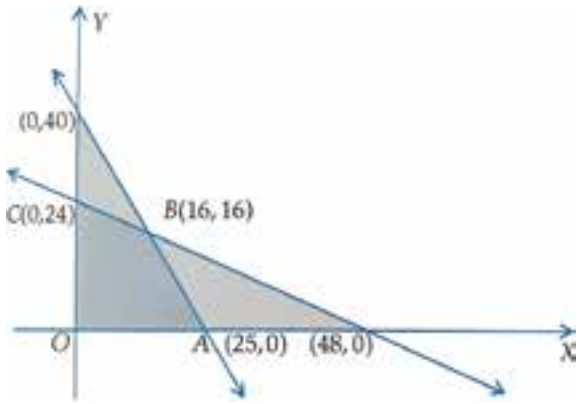
(a) x=a, x=b, the curve y=f (x), and the X-axis

(b) y=a, y=b, the curve x=f (y), and the Y-axis

(c) x=0, x=b, the line y=mx+c, and the Y-axis

(d) y=0, y=b, the curve x=f(y), and the X-axis

16. The maximum value of $z = 4x + 3y$, if the feasible region for an LPP is as shown in the figure, is to be determined.



(a) 100 (b) 72

(c) 112 (d) None of these

17. $+\int_{-2}^{2} \sin x\, dx + (2 + 2) = 4$ Which of the following function is decreasing on $(0, \pi/2)$?

(a) $\sin 2x$ (b) $\cos 3x$

(c) $\tan x$ (d) $\cos 2x$

18. The condition that $f(x) = ax^3 + bx^2 + cx + d$ has no extreme value is

(a) $b^2 > 3ac$ (b) $b^2 = 4ac$

(c) $b^2 = 3ac$ (d) $b^2 < 3ac$

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19. **Assertion :** If A is a matrix of order 2×2, then $|\text{adj A}| = |\text{A}|$

Reason : $|\text{A}| = |\text{A}^{\text{T}}|$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

20. Let us define $\tan^{-1}\text{A} + \tan^{-1}\text{B} = \tan^{-1}\left(\frac{\text{A}+\text{B}}{1-\text{AB}}\right)$

Assertion : The value of $\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{1}{7}\right)$ is $\frac{\pi}{4}$.

Reason: If $x > 0, y > 0$ than $\tan^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y-x}{y+x}\right) = \frac{\pi}{4}$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. If $\vec{a} = (2, 3, -5)$ and $\vec{b} = (2, 2, 2)$, then prove that $\vec{a}$ and $\vec{b}$ are mutually perpendicular.

22. Two dice are thrown. Find the probability that the numbers appearing have a sum 8 if it is known that the second die always exhibits 4.

23. The direction ratios of a straight line are 1,3,5. Find its direction cosines.

24. Solve – $\frac{dy}{dx} = e^{x-y}$

OR

Write down the order and the degree of the equation:

$$8x^2\frac{d^2y}{dx^2} - 7\left(\frac{dy}{dx}\right)^2 + 9 = 0$$

25. Find $\frac{dy}{dx}$ if $y = \cos^{-1}\frac{1-x^2}{1+x^2}$

OR

Find $\frac{dy}{dx}$ if $y = \sqrt{\sin x}$

SECTION - C

This section comprises of short answer-type questions (SA) of 3 marks each.

26. How fast is the volume of a ball changing with respect to its radius when radius is 3 m?

27. Is the function $f : R \to R$ onto function (where $f(x) = 2x$) Give reasons.

28. Prove, by Vector method, that the angle inscribed in a semi-circle is a right angle.

29. If $y = \tan^{-1}\left[\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right]$, prove that

$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}}$.

30. Prove that : $\sin^{-1}\frac{3}{5} - \cos^{-1}\frac{12}{13} = \sin^{-1}\frac{16}{65}$

OR

Find the value of $\sin\left(\sin^{-1}\frac{1}{2} + \cos^{-1}\frac{1}{2}\right)$.

31. If $A = \begin{bmatrix} 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix}$ and $B = \begin{bmatrix} 11 & 10 & 9 \\ 8 & 7 & 6 \end{bmatrix}$,

then find $A + B =$

OR

If $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then prove that

$f(x+y) = f(x).f(y)$

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Solve the following L.P. Problem graphically.

Maximise $Z = -x + 2y$

Subject to the constraints:

$$x \geq 3$$

$$x + y \geq 5$$

$$x + 2y \geq 6$$

$$y \geq 0$$

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33. Find the values of p, so that the lines

$$l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2} \text{ and}$$

$$l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$$

are perpendicular to each other. Also, find the equation of a line passing through a point $(3, 2, -4)$ and parallel to line l_1.

34. Solve : $\sqrt{a+x}\frac{dy}{dx} + x = 0$

OR

Solve : $\frac{dy}{dx} - \frac{xy}{1-x^2} = \frac{1}{1-x^2}$

35. Evaluate $\int \frac{3-5x}{2x^2+3x-2}dx$

OR

Prove that : $\int_0^{\pi/2} \log\sin x \, dx = -\frac{\pi}{2}\log 2$

SECTION - E

Case study based questions are compulsory.

36. A car carrier trailer, also known as a car-carrying trailer, car hauler, or auto transport trailer, is a type of trailer or semi-trailer designed to efficiently transport passenger vehicles via truck. Commercial-size car carrying trailers are commonly used to ship new cars from the manufacturer to auto dealerships. Modern car carrier trailers can be open or enclosed. Most commercial trailers have built-in ramps for loading and off-loading cars, as well as power hydraulics to raise and lower ramps for stand-alone accessibility.



A transport company uses three types of trucks T_1, T_2 and T_3 to transport three types of vehicles V_1, V_2 and V_3. The capacity of each truck in terms of three types of vehicles is given below:

	V_1	V_2	V_3
T_1	1	3	2
T_2	2	2	3
T_3	3	2	2

Using matrix method find:

(i) The number of trucks of each type required to transport 85, 105 and 110 vehicles of V_1, V_2 and V_3 types respectively.

(ii) Find the number of vehicles of each type which can be transported if company has 10, 20 and 30 trucks of each type respectively.

37. Due to the growing need to reduce emissions across the world, most countries are replacing existing fuel-based transportation with electric mobility. India, still in the very early stages of adopting EV's can be a huge market for E-bike rental companies. As most people are now aware of traffic congestion and pollution, India is looking towards a cleaner mode of transportation to tackle the situation.



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An owner of an electric bike rental company have determined if they charge customers ₹ x per day to rent a bike, where $50 \leq x \leq 200$ then number of bikes (n), they rent per day can be shown by linear function $n(x) = 2000 - 10x$. If they charge ₹ 50 per day or less, they will rent all their bikes. If they charges ₹ 200 or more per day they will not rent any bike.

Based on the above information, answer the following questions.

(i) Find the expression for total revenue R as a function of x.

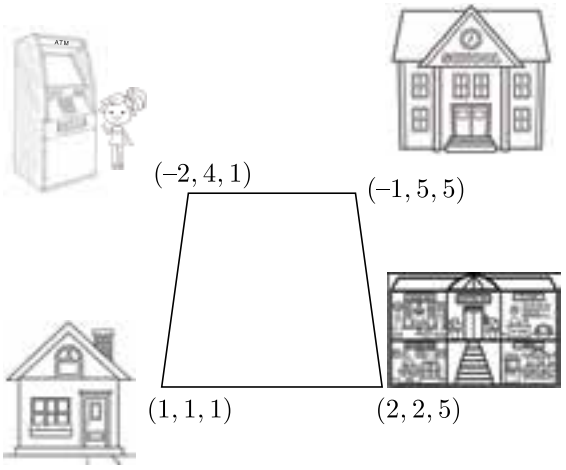
(ii) Find the value of x at maximum revenue.

(iii) What is the revenue collected by the company at $x = 260$? Find the number of bikes rented per day, if $x = 105$.

OR

(iv) Find the maximum revenue, collected by company.

38. Lavanya starts walking from his house to shopping mall. Instead of going to the mall directly, she first goes to ATM, from there to her daughter's school and then reaches the mall. In the diagram, using co-ordinate geometry the location of each place is given.



Based on the above information, answer the following questions.

(i) What is the distance between House and ATM?

(ii) What is the distance between ATM and school?

(iii) What is the total distance travelled by Lavanya?

OR

(iv) What is the extra distance travelled by Lavanya in reaching the shopping mall?

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